



GLOBAL FABRIC DAY 2026 - Toronto

Saturday, June 27, 2026



<https://globalfabric.community>

#GlobalFabricDay #GFD26



Global Fabric Day 2026 - Toronto



Data Loss Prevention for OneLake

Speaker: Mark Shoesmith

Solutions Architect



What we'll cover

- How Microsoft Fabric & Microsoft Purview work together.
- What can Microsoft Purview help with to prevent oversharing.
- Data Classification vs. Sensitivity Labels.
- Information Protection vs. Data Loss Prevention.
- Adoption Considerations.

Integrated solutions that address customer needs to safely drive business value and AI innovation



Challenges

Fragmentation

Evolving regulatory landscape

Growing data estates

Safeguarding for AI



What customers want



An Integrated analytics platform with security and governance solutions



Sophisticated security and governance solutions that adapt to regulations as they are released



Governance and security solutions that integrate across your entire, cross-cloud data estate



Trustworthy measures that provide assurance for AI innovation



Microsoft Fabric

The data platform for the era of AI

AI-powered data platform

Give your teams the AI-powered experiences needed for any data project in a pre-integrated and optimized SaaS environment

Open and AI-ready data lake

Access your entire multi-cloud data estate from a single data lake, work from the same copy of data, and ensure data is ready to power AI innovation

AI-enabled business users

Empower everyone to better understand data with AI-enhanced experiences and visuals embedded in the apps they use everyday

Mission critical foundation

Deploy and manage Fabric with category-leading performance, instant scalability, shared resilience, and built-in security, governance, and compliance

Microsoft Fabric

The unified data platform for AI transformation



Data Factory



Analytics



Databases



Real-Time
Intelligence



Power BI

Fabric Platform



AI



OneLake



Purview

What is data governance?

"Data Governance is defined as the exercise of authority and control (planning, monitoring, and enforcement) over the management of data assets."

Source: DAMA Data Management Body of Knowledge V2 (DMBOK2)

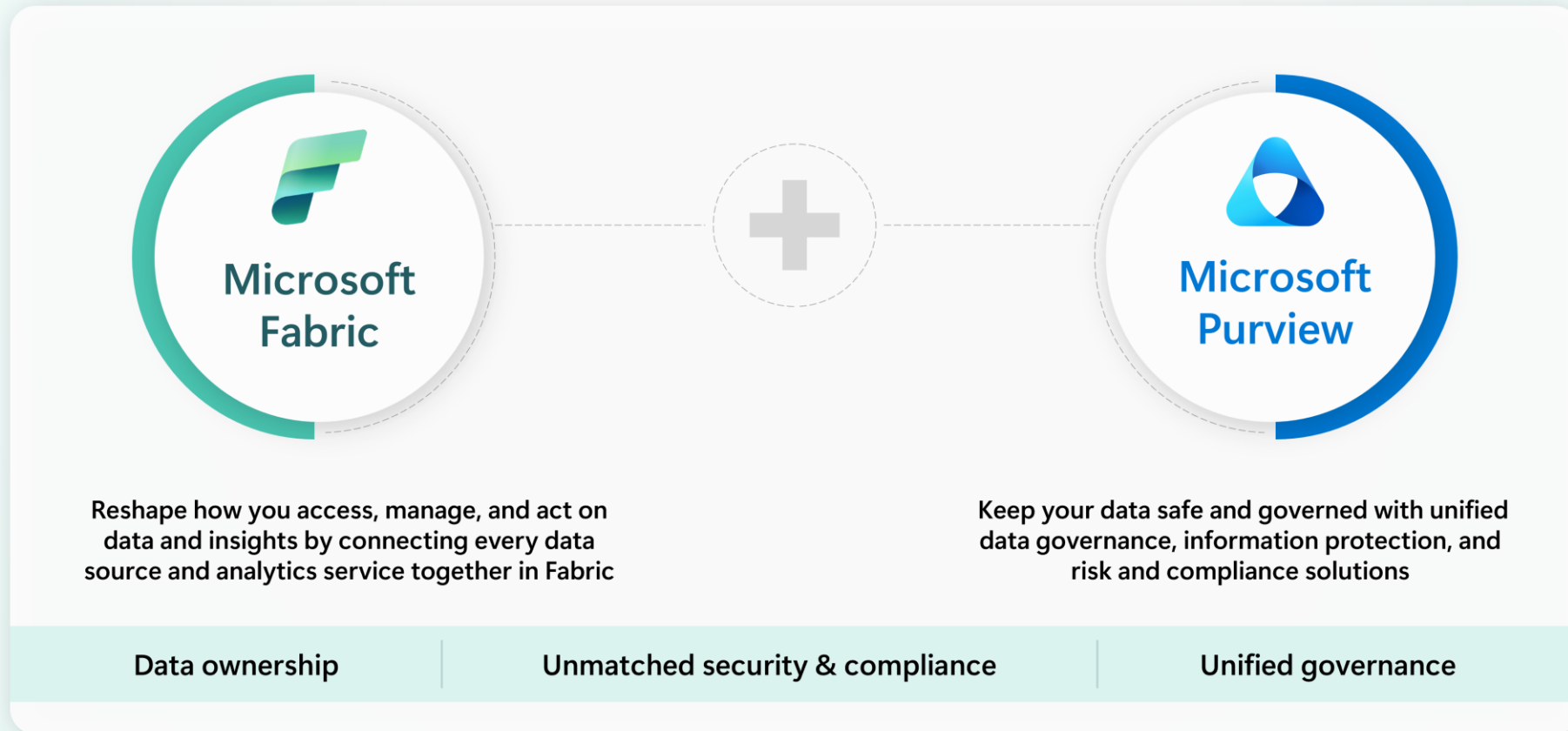
"Data governance encompasses the people, processes, and technology required to create a consistent and proper handling of an organization's data across the business enterprise."

Source: Wikipedia

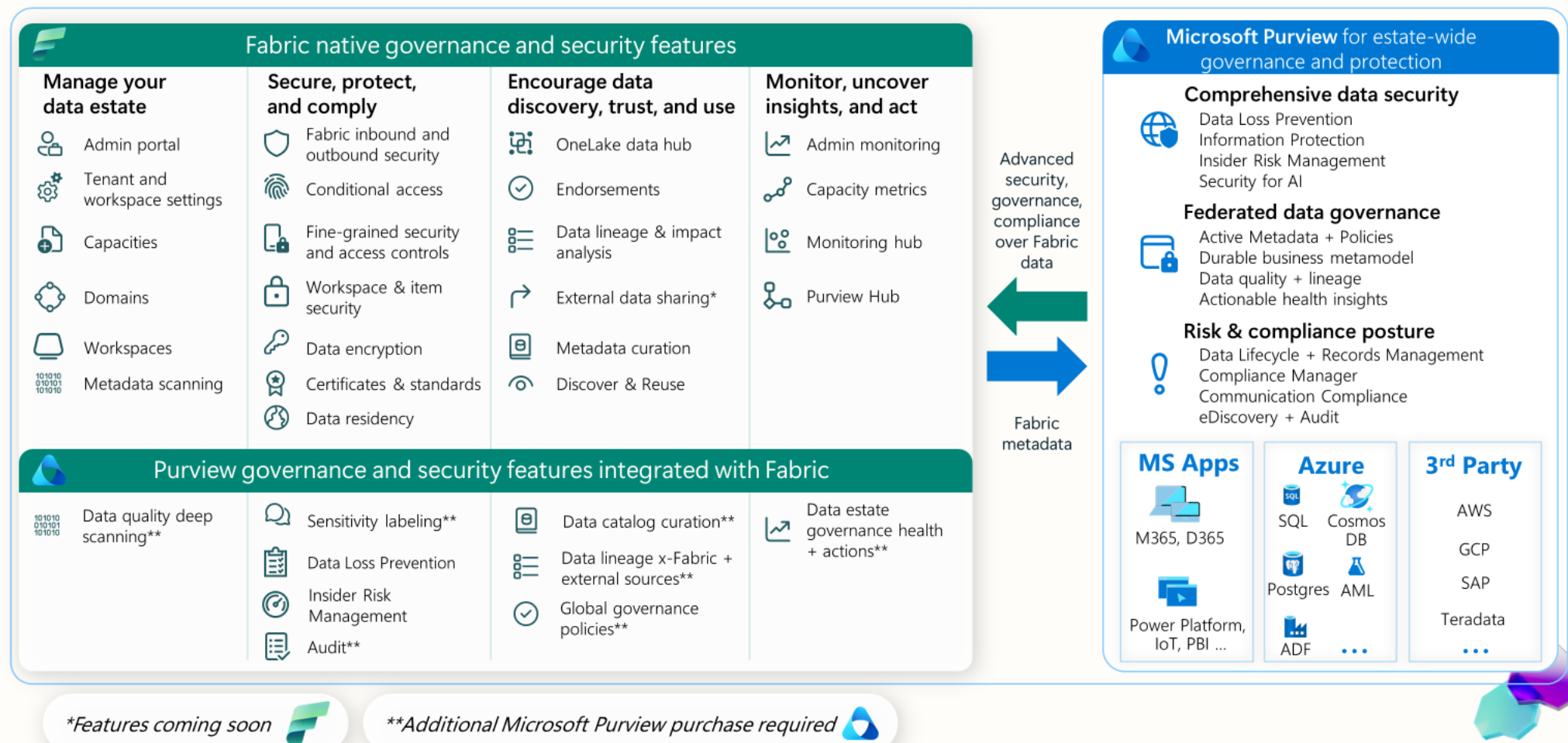
"Data governance is the orchestration of people, processes, policies and technology to formally manage and protect structured and unstructured data assets to guarantee commonly understood, trusted and secure data throughout the enterprise"

Source: Mike Ferguson, Intelligent Business Strategies

Unify analytics and governance with Microsoft Fabric & Microsoft Purview



Fabric Native Governance Features





Microsoft Purview

Integrated solutions to secure and govern the world's data



Data security

Secure data across its lifecycle,
wherever it lives

Data loss prevention
Insider risk management
Information protection



Data governance

Confidently activate your data
and accelerate time to insights

Unified catalog

Data discovery
Curation
Data quality
Data health



Data compliance

Manage critical risks and
regulatory requirements

Compliance manager
eDiscovery and audit
Communication compliance
Data lifecycle management
Records management

Unstructured and structured data

Traditional and AI generated data

Microsoft and multi-cloud

Shared platform capabilities

AI-based efficiency, Data map, Classification, Labels, Audit logs, Policies, Data connectors

Purview support across the data estate

Microsoft 365,
Devices & Copilots

Microsoft data estates, Copilots,
AI apps & Agents



Microsoft
Office



Microsoft
SharePoint



Microsoft
Exchange



Microsoft
Teams



Windows/
Azure VD



Windows



Microsoft
Edge



Microsoft
Fabric



Azure AI



Dataverse



Azure
Databases



Azure
Databricks



M365
Copilot



Security
Copilot



Copilot in
PowerBI



Azure
OpenAI



Copilot
Studio



Foundry

Classification | Labels | DLP | Adaptive Protection

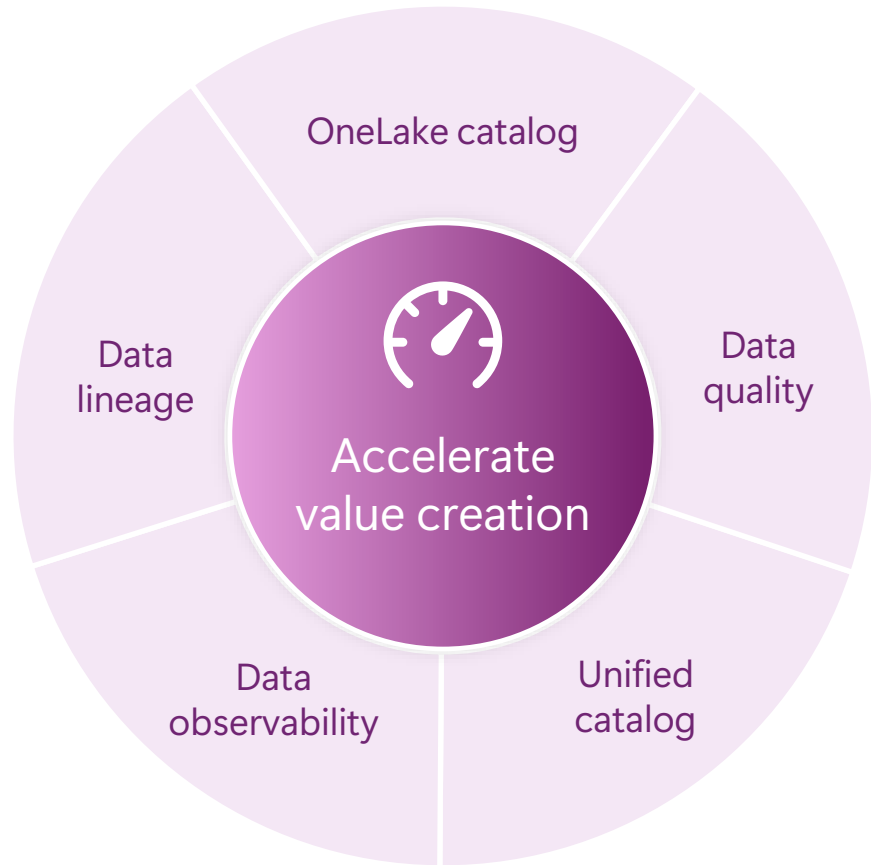
Shared capabilities

The right foundation for **trustworthy** data innovation

Seamlessly secure



Confidently activate



Security and compliance incidents come in many forms with your data and analytics

Concerns

Protect and prevent data loss with Fabric and Purview

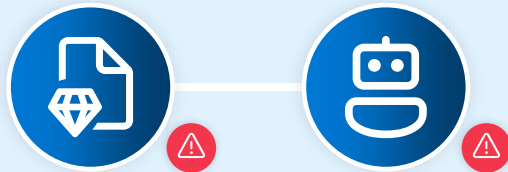
User accesses data that is confidential and should only be viewed by certain people



Data theft by malicious insider



User negligently overshares sensitive data and uploads data in Fabric

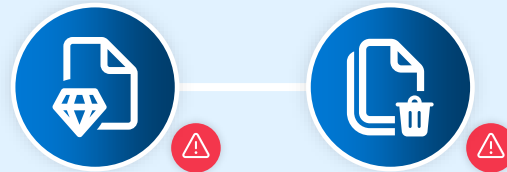


Data leak by negligent user



Discover + mitigate hidden risks to data

User deletes sensitive information in Fabric before leaving the organization

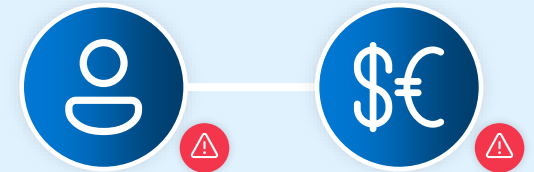


Data sabotage by disgruntled insider



Simplify compliance with your data

Users do not comply with regulations for their data



Data compliance fines are given to the organization



Defense in Depth

Microsoft Fabric security remains foundational, but Microsoft Purview adds information protection and compliance controls

Users, Apps, Copilot and Agents

DLP Policies Detect, alert, restrict where supported

Sensitivity Labels Business sensitivity & Protection policies

Classification Sensitive Information Types (SITs), classifiers, scans, metadata

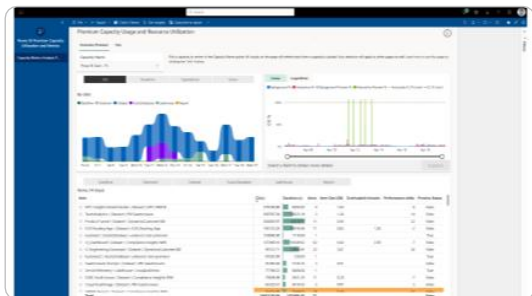
Fabric Security Workspace roles, item permissions, OneLake security

Data Estate OneLake, Lakehouse, Warehouse, Semantic Models, Reports

Important: DLP, and labels augment security. They do not replace Fabric RBAC, workspace roles, item permissions or OneLake security design.

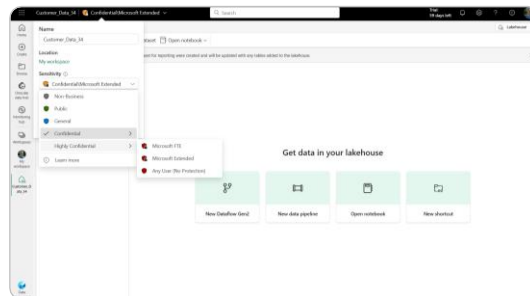
Protect and prevent data loss with Fabric and Purview

Fabric and Purview: Foundational and advanced security options



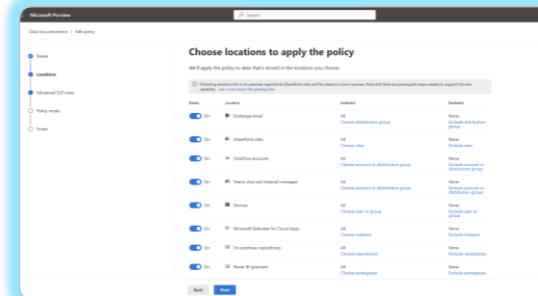
Manage security across Fabric

With [OneLake security](#), you can define access permissions once and Fabric will enforce it consistently across all engines



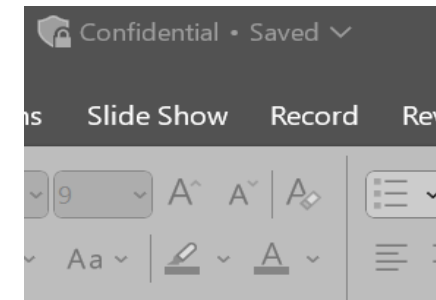
Apply advanced protection to your data

[Microsoft Purview Information Protection](#) offers advanced security the form of sensitivity labels to identify sensitive data consistently



Prevent data oversharing

[Microsoft Purview Data Loss Prevention](#) prevents data oversharing in Fabric through forms of set policies and ensures sensitive data (PII, credit card numbers, etc.) does not end up in the wrong locations



Consistent classification in M365 and analytics

Classify and label sensitive data consistently across your digital estate, including Microsoft 365 apps, Microsoft Fabric, and beyond

Related Features

Data Catalog, labelling, DLP, audit, and insider risk

Data Governance

Data Map, Unified Catalog, metadata scanning, lineage, business domains, data products, and data curation.

Information Protection

Sensitivity labels, and protection policies that classify and protect Fabric / Power BI content with Purview labels.

Data Security

DLP, audit, and related controls to detect risky movement or usage of sensitive data and support investigation.

- Classification is primarily a discovery, and understanding mechanism.
- Sensitivity labels represent the organization's information taxonomy.
- Microsoft Purview protection access control policies (protection policies) enable organizations to control access to items in Fabric using sensitivity labels.
- **Microsoft Purview Data Loss Prevention (DLP)** policies evaluate content against labels and/or detections, and then acts according to policy.

Classification: What Data Do We Have?

Microsoft Purview Data Map categorizes assets using logical classes

Microsoft definition

Classification is a way of categorizing data assets by assigning unique logical tags, or classes. It helps make data easier to understand, search, and govern.

Common detections

Passport Number • Driver's Licence Number • Credit Card Number • SWIFT Code • Person's Name • Email • other system or custom classifications

Data element	Classification outcome	Governance meaning
credit_card_number	Credit Card Number	Regulated financial identifier
employee_sin	National identifier	Personal/confidential information
customer_email	Email Address	PII / contact information
project_code	Custom classification	Business-specific governance signal

Sensitivity Labels: How Sensitive Is It?

Labels express business sensitivity and can apply protection behaviour

Public

Internal

Confidential

Highly Confidential

Restricted

Microsoft Fabric Information Protection

Microsoft Fabric, and Power BI categorize, and protect sensitive data using sensitivity labels, and policies from Microsoft Purview Information Protection.

Access control nuance

Labels can be associated with Microsoft Purview protection policies. Where configured, and supported, those protection policies can control access to labelled Fabric / Power BI content.

Labels are not just metadata only. In mature Microsoft Information Protection (MIP) implementations, labels can drive user experience, policy decisions, encryption / usage restrictions and audit context, subject to workload support and configuration.

Information protection in Fabric and Power BI enables organizations to categorize, and protect their sensitive data using sensitivity labels and policies from Microsoft Purview Information Protection. In addition, Fabric, and Power BI provide additional capabilities to help organizations achieve maximum sensitivity label coverage, and to manage, and govern their sensitive data.

Classification vs. Sensitivity Labels

Similar, but not the same

Dimension	Classification	Sensitivity Label
Primary question	What type of data is this?	How sensitive is it and how should it be handled?
Typical examples	Credit card, passport, name, email	Public, Internal, Confidential, Highly Confidential
Main purpose	Discovery, search, governance insight, risk context	Business sensitivity, protection, policy enforcement signal
Applied by	Purview scans, classifiers, manual edits	Users, policies, inheritance, APIs / admin configuration
Used by DLP?	Yes, as sensitive information detections / Sensitive Information Types (SITs)	Yes, as label-based DLP conditions

How Sensitivity Labels Flow Through Fabric

Inheritance helps protect data as it moves from source to consumption



Documented inheritance patterns

Downstream inheritance for Power BI semantic models and reports vs. other Fabric items varies, and distinguishes between downstream inheritance, from inheritance upon item creation.

Practical design point

Use label inheritance to reduce unlabeled downstream artefacts, but validate exact behaviour per workload, tenant setting, and label policy.

<https://learn.microsoft.com/en-us/fabric/governance/information-protection#capabilities>

Information Protection vs. Data Loss Prevention

Classify once, enforce everywhere

Microsoft Purview Information Protection

- Allows you to discover, classify, and protect Fabric data using sensitivity labels.
- Sensitivity labels can be set on all Fabric items.
- When a protection policy is associated with a label E.g., 'Highly Confidential' only the users, and groups specified in the policy retain access; Everyone else is blocked automatically, at runtime.
- Unlike traditional workspace-level permissions, protection policies follow the data's classification.
- If an item is labeled 'Highly Confidential – Finance', only the finance team retains access, regardless of which workspace the item lives in.
- Protection policies currently support Power BI semantic models, but not reports or dashboards directly.
- Data remains protected when it's exported via supported export paths.
- Compliance admins can monitor activities on sensitivity labels in Microsoft Purview Audit.

Information Protection vs. **Data Loss Prevention**

Inheritance helps protect data as it moves from source to consumption

Microsoft Purview Data Loss Prevention (DLP)

- DLP policies support structured data in Fabric, such as lakehouses, warehouses, various databases, and semantic models.
- DLP policies detect upload of sensitive data into OneLake and can also enforce access restrictions on sensitive assets across Fabric KQL Database, Fabric SQL Database, and Fabric Data Warehouse.
- Policies evaluate sensitivity labels, and a subset of sensitive info types (such as credit card, and social security numbers i.e. classifications) on structured data in these stores, to limit user actions when data is classified as sensitive.
- DLP policies can be configured to generate policy tips for Fabric users, and alerts for security admins.
- DLP policies can be configured to restrict access to the item (preview).
- When you configure a policy with the restrict access action and a policy match occurs, the policy restricts access to the data owners, or to members of the organization, depending on the policy configuration. All other users lose access to the item.
- DLP policies can also be configured to allow workspace admins to override them.

What DLP for Fabric can inspect today

Lakehouses

Delta tables

Warehouses

SQL tables

Semantic Models

Import mode

KQL Databases

Real-time

SQL databases

Fabric SQL

Mirrored Databases

Replicated

Cosmos databases

NoSQL

Workspace must sit on Fabric Capacity

Trial capacities count

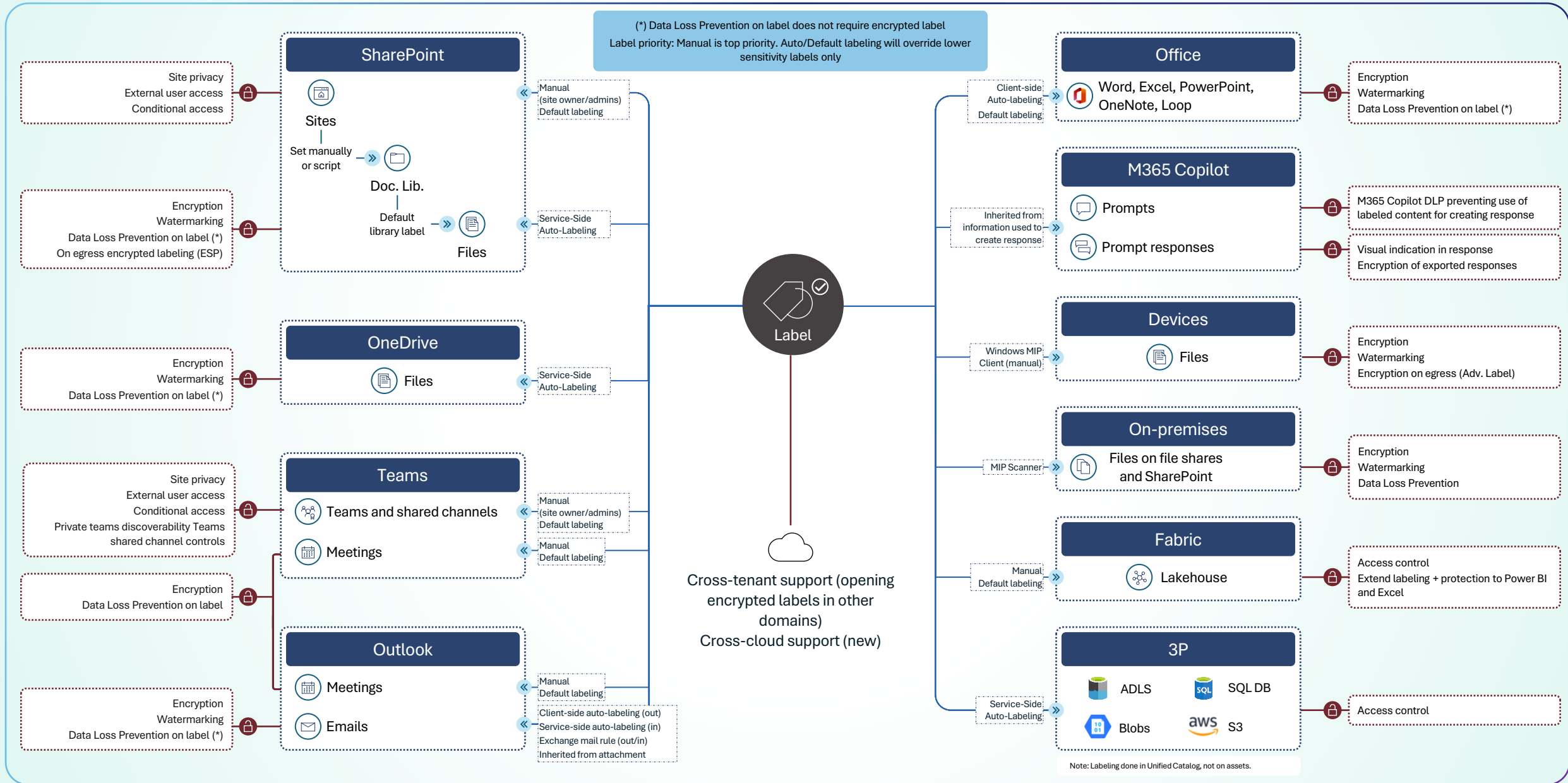
Delta format only. DLP for Fabric inspects tables stored as Delta (treated as text files for scanning). Several Delta Parquet types are out of scope (Binary, Struct, Array, Map, JSON, Interval, timestamp_ntz, and LZ4/Zstd/Gzip-compressed data).

No policy templates for Fabric ; always choose the custom policy path.

Microsoft Purview Labelling & Protection

Legend

— Labeling methods
— Protection methods



What Should Happen When Risk Is Detected?

DLP evaluates sensitive content against policy and triggers actions

Conditions

A DLP rule can detect sensitivity labels and / or sensitive information types configured in the policy.

Evaluation

Fabric evaluates supported items when relevant data changes occur; semantic models are also evaluated in publish / refresh scenarios according to your internal DLP material.

Actions

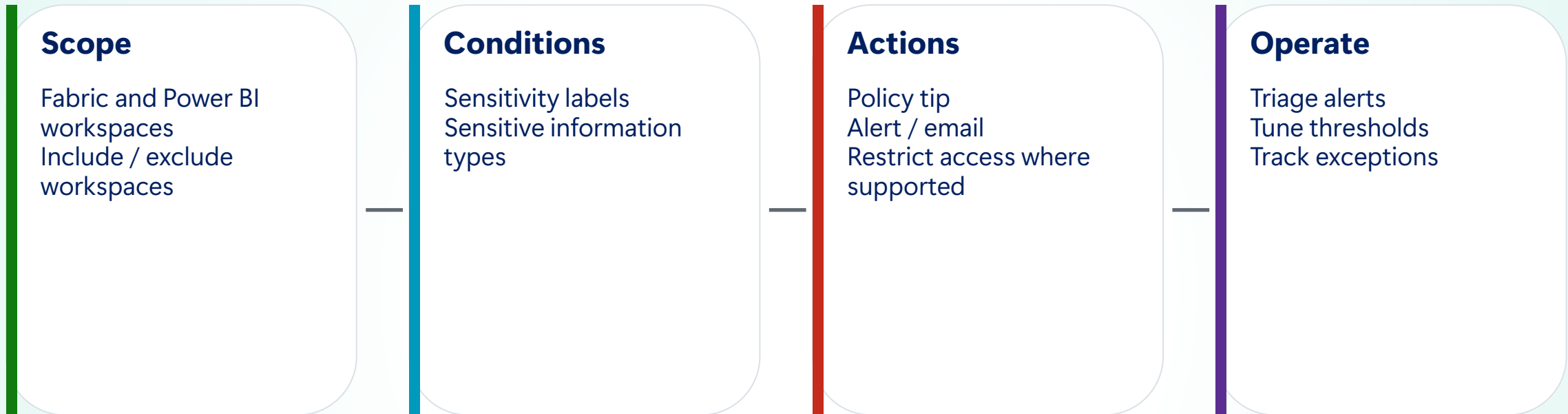
Policy tips, alerts / email notifications, incident investigation and access restrictions where the configured action is supported.



DLP is the policy execution layer it uses what Purview knows about the content, and what MIP labels say about handling rules.

DLP Policy Design Pattern

A repeatable structure for Fabric / OneLake DLP rules



Recommended rollout start in simulation / monitoring, tune false positives, and stakeholder routing, then introduce stronger actions for confirmed high-risk scenarios.

How a Fabric DLP policy actually works

Define

In the Microsoft Purview portal, you create a custom policy scoped to Fabric workspaces, with conditions and actions.

Evaluate

Fabric evaluates an item when its data changes, or, for semantic models, on publish & refresh.

Match

Content is checked against sensitivity labels and sensitive info types (SITs) in the policy.

Act

On a match, the policy fires its actions: policy tip, alert/email, and/or restrict access.

What a policy can match on, and what it cannot

✓ Supported conditions

Sensitivity labels

Key off the label IP already applied (e.g. Highly Confidential).

Sensitive info types (SITs)

Built-in patterns with confidence levels e.g. Credit Card Number, Canada Social Insurance Number, bank account.

Combine & tune

Set instance counts and confidence to control noise.

✗ Not supported (silently returns nothing)

Advanced classifiers are ignored:

- Exact Data Match (EDM)
- Trainable classifiers
- Credential classifiers
- Named entities

Be Careful: if a policy condition uses one of these, that condition returns no results even when the data is present, other classifiers in the same policy still work, so the policy looks 'live' while quietly missing matches.

Three possible actions

DETECT

Policy tip

Marks the item with an icon and an in-product explanation for the data owner. Notification only.

DETECT

Alert / email

Records an alert on the DLP Alerts page in Purview and can email admins and named users.

ENFORCE

PREVIEW

Restrict access

Removes access for everyone except data owners or org members, per policy. The only action that changes who can reach the data.

So today: policy tips + alerts are your GA monitoring; restrict access is your only enforcement lever, and it's in preview.

When does evaluation actually run?

Semantic models

Evaluated on:

- Publish
- Republish
- On-demand refresh
- Scheduled refresh

Not evaluated for DirectQuery / live connection or mixed-storage models.

Fabric items (lakehouse, warehouse, DBs)

Evaluated when the data changes:

- Getting new data
- Connecting a new source
- Adding tables
- Updating existing tables

Event-driven: an untouched item isn't re-scanned until it changes again.

Current Limitations

Custom policy only

No Fabric DLP templates start from the custom policy option every time.

Delta-only scanning

Non-Delta tables and several Parquet types (Struct, Array, Map, JSON...) aren't scanned.

No DirectQuery / live

Those semantic models are never evaluated, only Import-mode content is inspected.

Service Principals (SPN)

SPN-owned or SPN-triggered publish/refresh skips evaluation entirely.

Advanced classifiers ignored

EDM, trainable, credential, named-entity conditions return nothing - silently.

Detection-first

Restrict access is preview; there's no block-share / block-export action today.

Why This Matters for Copilot and AI

Labels and DLP become increasingly important as data is activated through natural language

Governance challenge

Fabric makes data easier to activate across analytics and AI experiences. That increases the importance of consistent classification, labelling and oversharing controls.

Control model

Permissions determine what users can access. Labels and DLP help determine how sensitive data should be handled, monitored and protected when it is used or moved.



AI governance is not a separate control plane it depends on high-quality permissions, labels, DLP, auditing and data ownership.

Decision Tree

Which feature should I use?

Question

Need to find sensitive data?

Use classification, classifiers, scans and Data Map.

Need to express handling rules?

Use sensitivity labels and protection policies.

Need to detect risky movement?

Use DLP policies with labels and/or sensitive information conditions.

Need to control who can access?

Use Fabric security, OneLake security, RLS/OLS/CLS and item/workspace permissions.

Need to investigate?

Use audit, DLP alerts, activity explorer and security operations processes.

Prerequisites & Roles

Use this as a readiness checkpoint before a PoV or implementation

Licensing

DLP and advanced information protection require appropriate Microsoft 365 / Purview compliance licensing. Fabric / Power BI licences and capacity requirements also apply.

Tenant setup

Sensitivity labels must be defined and published to relevant users / groups. Fabric and Purview need tenant alignment for native governance integration scenarios.

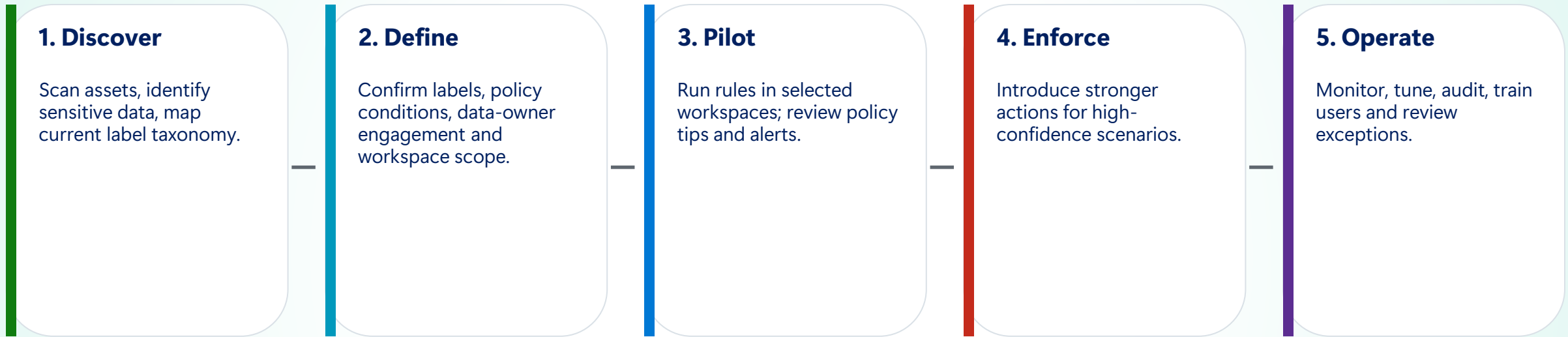
Roles

Typical roles include Fabric / Power BI admin, Purview admin, compliance / information protection admin, data owner and workspace owner.

'Do you already have a Microsoft Information Protection label taxonomy used for M365, and is it published to the Fabric user groups?'

Recommended Rollout Roadmap

A pragmatic implementation model for Fabric + Purview protection



Delivery recommendation: separate information architecture decisions (taxonomy) from enforcement decisions (DLP actions) to avoid over-blocking early adopters.

Licensing, capacities & what it costs to run

Licensing

Purview DLP for Fabric & Power BI rides on Microsoft 365 E5 / E5 Compliance entitlements. Authoring needs a Purview compliance role.

Capacity

Only workspaces on Fabric or Premium capacity are in scope. Fabric trial capacity works in the lab.

Billing

DLP evaluation consumes capacity (CUs) as it's a metered background workload. Factor it into capacity sizing for busy lakehouses.

Roles you'll need to author & monitor

Author policies: Compliance Admin · Compliance Data Admin · Information Protection (Admin) · Security Admin.

View activity & alerts: Compliance Admin · Security Admin · Compliance Data Admin (Activity Explorer).

Plus, in Fabric: a Fabric Admin to confirm capacity, and workspace Admin/Member to build the demo items.

Monitoring & the data owner response loop

Alerts page

Triage matches in the Purview DLP Alerts page: severity, item, policy, and the matched info types in one queue.

Activity Explorer

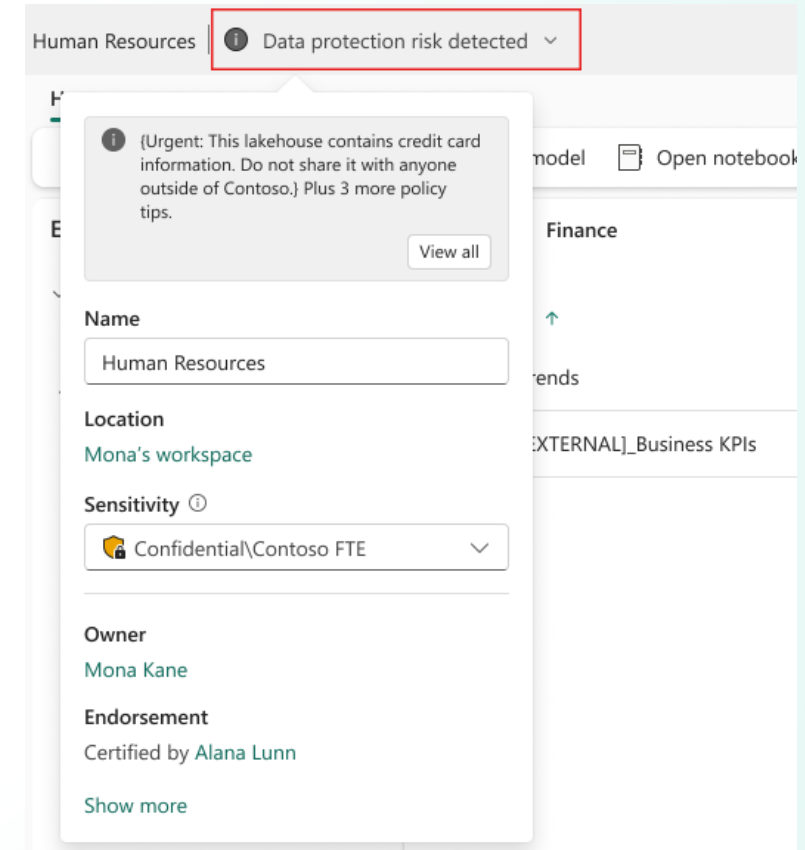
Trend DLP activity across Fabric & Power BI for tuning: needs a compliance/security role to view.

Email + audit

Route emails to admins and named owners: correlate with Purview Audit for the full activity trail.

What the data owner sees and does

A DLP icon on the item → hover card → **“View all” side panel** with the policy details and guidance. They remediate (remove/relabel/secure the data), and note a hidden tip isn't deleted; it returns on the next visit until the match is resolved.



Operating Model: Who Owns What?

Clarify responsibilities so controls are sustainable

Role	Primary responsibilities
Compliance / Info Protection Admin	Maintain labels, label policies, DLP policies and alert routing.
Fabric Admin	Tenant settings, workspace governance, capacity alignment and platform controls.
Data Owner	Confirm sensitivity, approve usage, respond to policy tips and remediation requests.
Security / SOC	Investigate alerts, identify risky movement and coordinate incident response.
Delivery Team	Apply patterns consistently in lakehouse, warehouse, semantic model and report design.

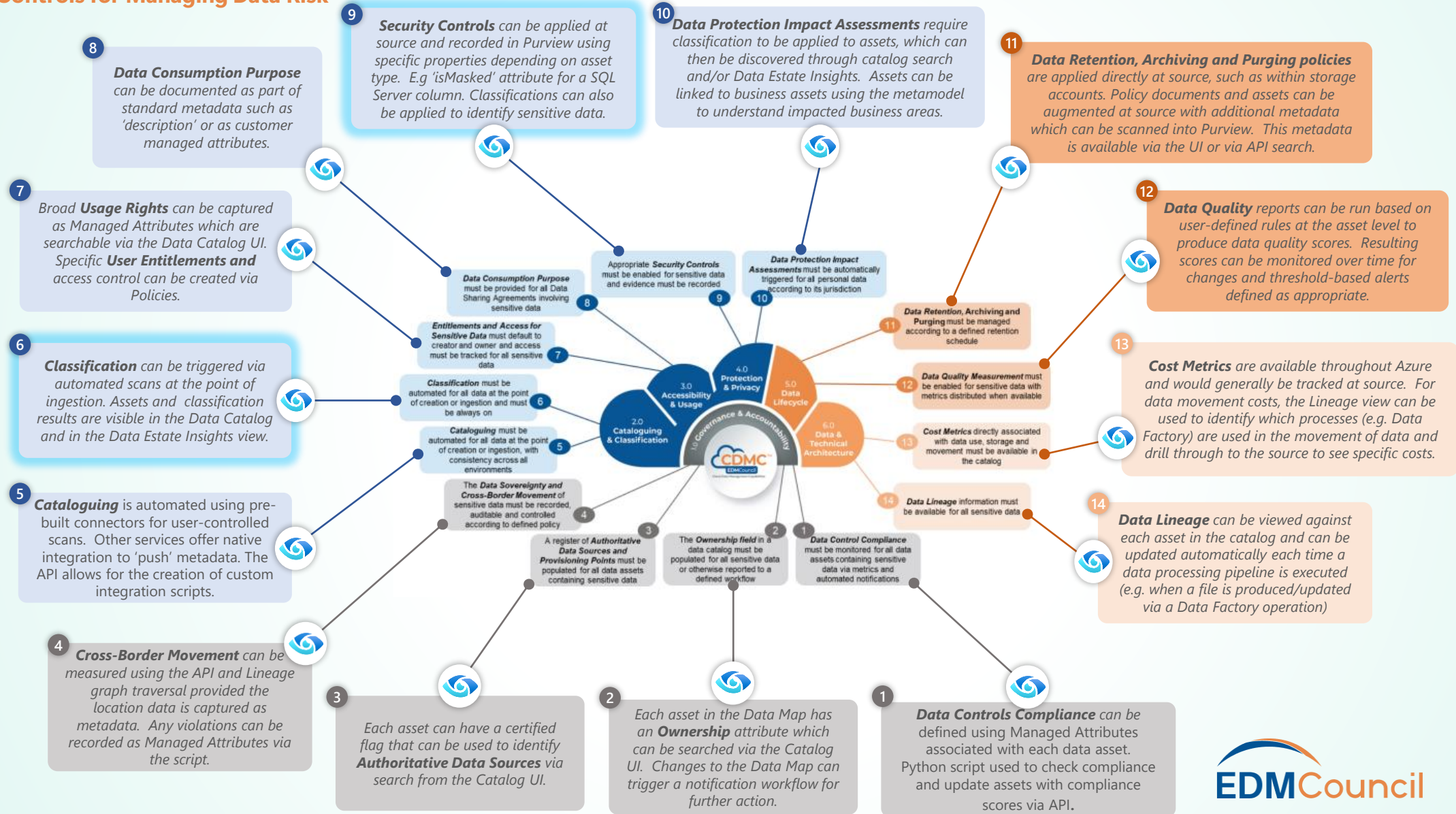
CDMC™ Cloud Data Management Capabilities



- A comprehensive assessment, and certification framework, encompassing business, operations, and technology best practices for data management in cloud and hybrid-cloud environments.
- Launched in September 2021, CDMC™ was developed by the EDM Council's CDMC™ Workgroup, with collaborative participation from 300+ professionals from 100+ firms cross-industry.
- Microsoft joined the working group at the outset, when beginning its journey building out what is now known as Microsoft Purview.
- Microsoft Purview allows client companies across all sectors to implement best practices within their own operational environments, and ensure that the 14 Key Controls will protect their sensitive data cross-jurisdiction, and speed their own certification against the CDMC Key Controls.

CDMC: Cloud Data Management Capabilities

14 Key Controls for Managing Data Risk



Key Takeaways

Complimentary Capabilities

Classification discovers

It identifies and categorizes sensitive data so governance teams understand the estate.

Labels classify, and protect

They express business sensitivity, and can carry protection intent across supported experiences.

DLP enforces policy

It evaluates content against labels / detections, and acts through policy tips, alerts, incidents and restrictions.



Microsoft Fabric gives you the analytics platform.



Microsoft Purview gives you the information governance, and data security control plane needed to use it responsibly at scale.

Questions?



Thanks!

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References

Data classification in Microsoft Purview Data Map

<https://learn.microsoft.com/en-us/purview/data-map-classification>

Information protection in Microsoft Fabric

<https://learn.microsoft.com/en-us/fabric/governance/information-protection>

Sensitivity label downstream inheritance in Power BI

<https://learn.microsoft.com/en-us/fabric/governance/service-security-sensitivity-label-downstream-inheritance>

Configure DLP policies for Microsoft Fabric

<https://learn.microsoft.com/en-us/fabric/governance/data-loss-prevention-configure>

Data Map

<https://learn.microsoft.com/en-us/purview/data-map>

Note: Fabric, Purview and DLP capabilities evolve quickly. Validate supported item types, actions, licensing, and tenant settings against published features, and roadmap **before** committing scope in an SOW, or a production rollout.